

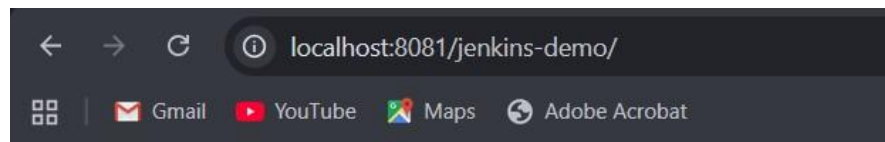
Semester	B.E. Semester VII – CMPN	
Subject	DevOps	
Subject Professor In-charge	Dr. Snehal Andhare	
Assisting Teachers	Prof. Snehal Andhare	
Classroom	M414A	
Student Name	Pratik Pujari	
Roll Number	23102B0008	
Grade and Subject Teacher's Signature		
Practical Number	4	
Practical Title	To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat or nginx server.	
Description	The practical demonstrates the creation of a Jenkins pipeline using Maven, Gradle, or Ant for automating the build, testing, and deployment of an application. The application is tested and deployed on a Tomcat server through the Jenkins pipeline, demonstrating the use of Continuous Integration and automated deployment.	
Program		

Output

Manual Test:

```
[INFO] -----
[INFO] Running com.vit.demo.CalculatorTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.079 s -- in com.vit.demo.CalculatorTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ jenkins-tomcat-demo ---
[INFO] skip non existing resourceDirectory C:\Users\thele\OneDrive\Desktop\Devops4\jenkins-tomcat-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ jenkins-tomcat-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ jenkins-tomcat-demo ---
[INFO] skip non existing resourceDirectory C:\Users\thele\OneDrive\Desktop\Devops4\jenkins-tomcat-demo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ jenkins-tomcat-demo ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug release 21] to target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ jenkins-tomcat-demo ---
[INFO] Skipping execution of surefire because it has already been run for this configuration
[INFO]
[INFO] --- war:3.5.1:war (default-war) @ jenkins-tomcat-demo ---
[INFO] Packaging webapp
[INFO] Assembling webapp [jenkins-tomcat-demo] in [C:\Users\thele\OneDrive\Desktop\Devops4\jenkins-tomcat-demo\target\jenkins-demo]
[INFO] Processing war project
[INFO] Copying webapp resources [C:\Users\thele\OneDrive\Desktop\Devops4\jenkins-tomcat-demo\src\main\webapp]
[INFO] Building war: C:\Users\thele\OneDrive\Desktop\Devops4\jenkins-tomcat-demo\target\jenkins-demo.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.776 s
[INFO] Finished at: 2026-08-19T22:36:40+05:30
[INFO] -----
```

Tomcat Application:



Jenkins CI/CD Pipeline Successful!

Application deployed on Apache Tomcat.

Build Tool: Maven

CI/CD Tool: Jenkins

Application Server: Apache Tomcat

Jenkins Pipeline Build:

	 Jenkins Jenkins-Tomcat-Pipeline #4 Search Share Help Status Changes Console Output Delete build #4 Git Build Data See Fingerprints Tests Pipeline Overview Edit build information Restart from Stage Replay Pipeline Steps Workspaces Previous Build ✓ #4 (Aug 19, 2026, 10:28:02 PM) Build Artifacts  jenkins-demo.war 2.85 KB view Started by user Talha Patrawala Revision 0ef5d68736475c5586739b846ef6a0755ab11be6d Repository https://github.com/Talha905/jenkins-tomcat-demo.git • refs/remotes/origin/main  Tests (no failures)  No changes. Add description Keep this build forever Started 12 min ago Took 25 sec REST API Jenkins 2.568.2
Conclusion	Thus, the practical provided an understanding of creating and executing a Jenkins pipeline for building, testing, and deploying an application. It demonstrated how Jenkins can automate the software delivery process and integrate build tools with application servers such as Tomcat, improving efficiency, and reducing manual effort.